

Browsing History

Viewing the history

```
git log --stat      # Shows the list of modified files
git log --patch     # Shows the actual changes (patches)
```

Filtering the history

```
git log -3          # Shows the last 3 entries
git log --author="Mosh"
git log --before="2020-08-17"
git log --after="one week ago"
git log --grep="GUI"    # Commits with "GUI" in their message
git log -S"GUI"        # Commits with "GUI" in their patches
git log hash1..hash2   # Range of commits
git log file.txt       # Commits that touched file.txt
```

Formatting the log output

```
git log --pretty=format:"%an committed %H"
```

Creating an alias

```
git config --global alias.lg "log --oneline"
```

Viewing a commit

```
git show HEAD~2
git show HEAD~2:file1.txt # Shows the version of file stored in this
commit
```

Comparing commits

```
git diff HEAD~2 HEAD    # Shows the changes between two commits
```

```
git diff HEAD~2 HEAD file.txt # Changes to file.txt only
```

Checking out a commit

```
git checkout dad47ed    # Checks out the given commit
git checkout master     # Checks out the master branch
```

Finding a bad commit

```
git bisect start
git bisect bad          # Marks the current commit as a bad commit
git bisect good ca49180 # Marks the given commit as a good commit
git bisect reset        # Terminates the bisect session
```

Finding contributors

```
git shortlog
```

Viewing the history of a file

```
git log file.txt        # Shows the commits that touched file.txt
git log --stat file.txt # Shows statistics (the number of changes)
for file.txt
git log --patch file.txt # Shows the patches (changes) applied to
file.txt
```

Finding the author of lines

```
git blame file.txt      # Shows the author of each line in file.txt
```

Tagging

```
git tag v1.0           # Tags the last commit as v1.0
git tag v1.0 5e7a828    # Tags an earlier commit
git tag                # Lists all the tags
```

```
git tag -d v1.0      # Deletes the given tag
```